Game Theory / Human Behaviour / Artificial Intellegence



THE PRISONER'S DILEMMA

THE SCIENCE OF COOPERATION

Can two smart decisions lead to a bad result for everyone?

Kateryna Soloviova

COOPERATION - BETRIRAYAL - CODE

WHAT IS GAME THEORY ?

The study of decisions where the result depends on what other people do.



GAME THEORY

The study of strategic desicion

The Prisonner's Dilemma

Mathematical model developed by Merrill Flood and Melvin Dresher around 1950

The Story

Albert Tucker introduced the famous story of two prisoners

TWO PRISONERS * NO COMMUNICATION

MY CHOICE / OTHER	C COOPERATE	D BETRAY
C COOPERATE	3 / 3	0 / 5
D BETRAY	5 / 0	1 / 1

Dominant Strategy : Betray

Nash Equilibrium: D/D 1 point each

The paradox: two smart individual decisions create a worse result for both.

CAN COOPERATION BECOME A WINNING STRATEGY?

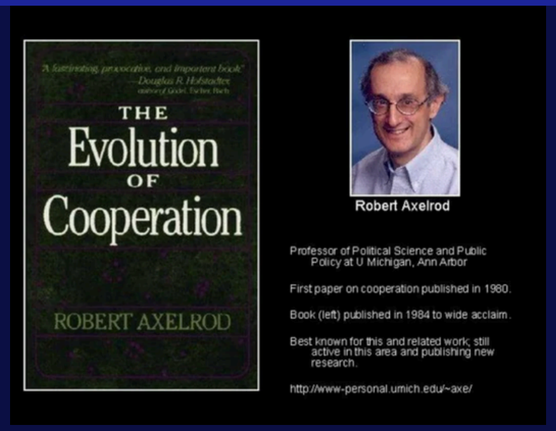
When players meet again, today's choice affects tomorrow.



1914

The Christmas Truce

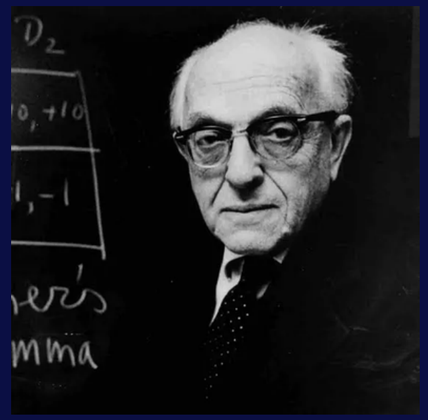
Soldiers on opposing sides temporarily stopped fighting in some areas



1980s

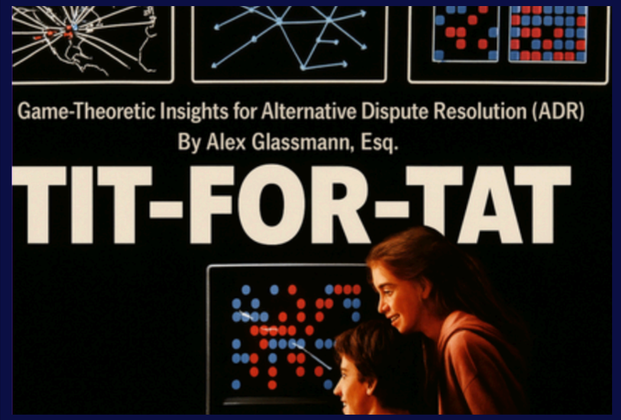
Robert Axelrod

Computer tournaments: different strategies played against each other.



Anatol Rapoport

Submitted Tit for Tat Strategy

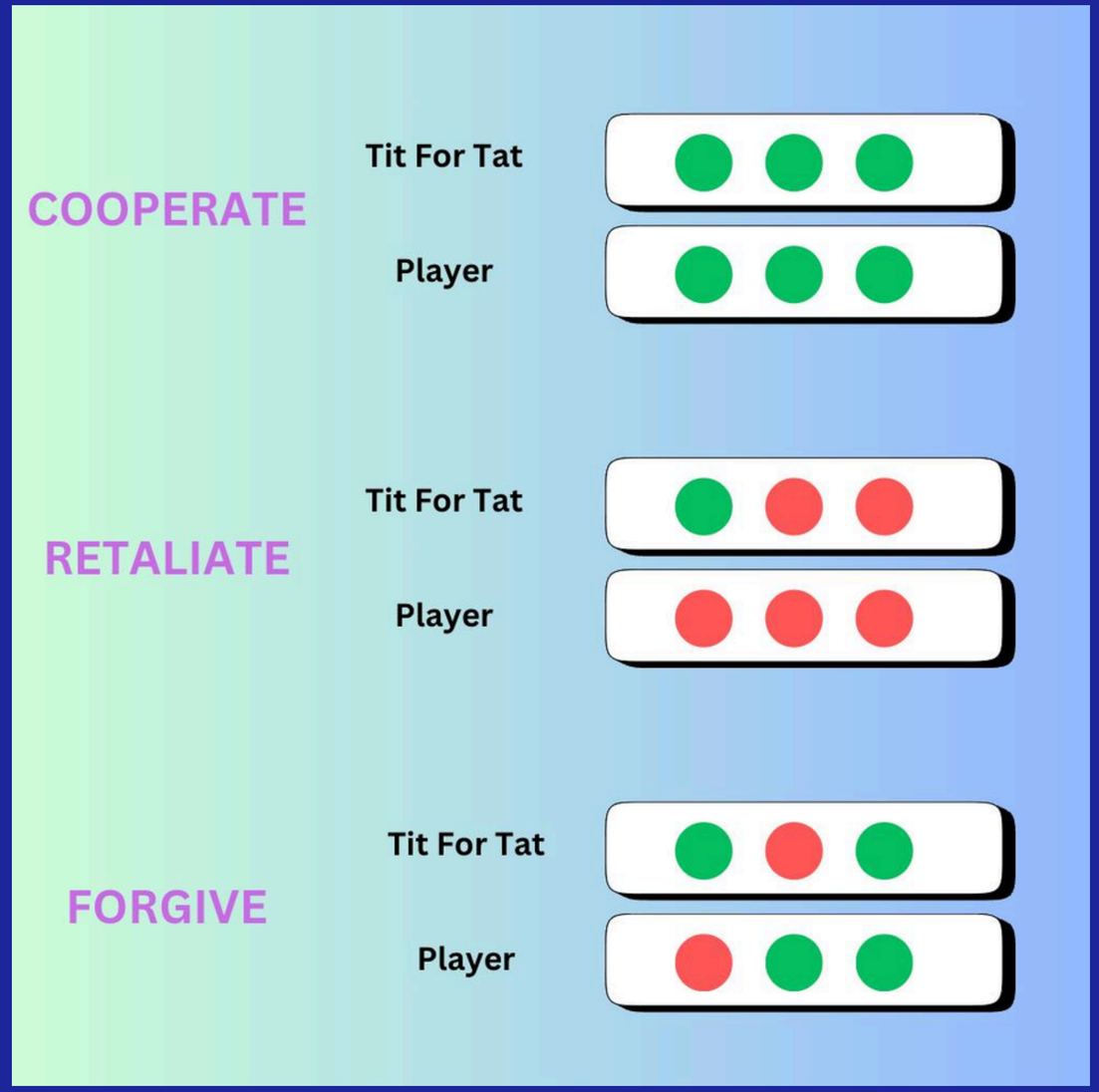


TIT FOR TAT

Cooperate first.
Then copy the last move

HOW DOES THE ‘TIT FOR TAT’ WORK?

The rule is simple: cooperate first, then copy the opponent’s last move



PYTHON

```
def tit_for_tat(opponent_moves):
    if not opponent_moves:
        return "C"

    return opponent_moves[-1]
```

C = cooperate
opponent[-1] = the last move

NOISE:
one mistake can start endless revenge

Generous Tit for Tat:
sometimes forgive

WHERE DOES COOPERATION WORK IN REAL LIFE?

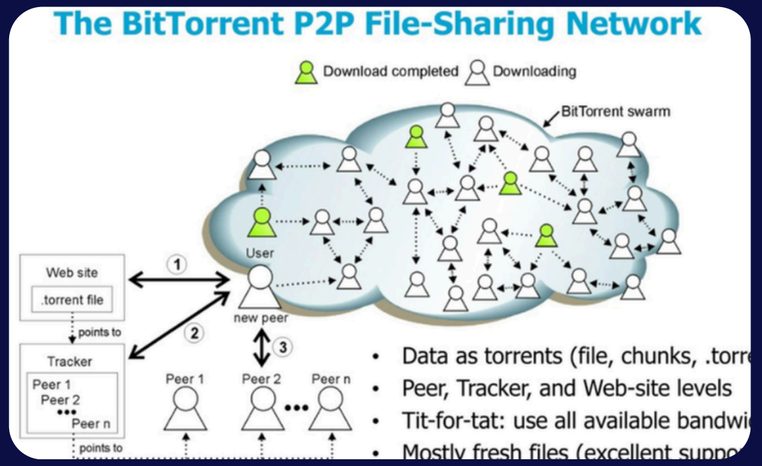
The same principle of mutual help appears in nature and technology.

BIOLOGY



Vampire bats share food with other bats.
They help each other because they may need help in the future.

TECHNOLOGY



BitTorrent is a file-sharing system.
Users share parts of files with each other. It is based on mutual contribution.

ARTIFICIAL INTELLIGENCE



Modeling how autonomous agents choose to cooperate or compete based on others' previous actions.



SOMETIMES, THE SMARTEST STRATEGY ISN'T TO WIN AGAINST EVERYONE.
IT'S TO LEARN HOW TO COOPERATE.

THANK YOU FOR YOUR ATTENTION!